

Domain and Range

ANSWER KEY



Domain from equations

- 1 State the domain of each function:

a $f(x) = \frac{1}{x-4}$ All real x with $x \neq 4$.

b $g(x) = \sqrt{2x-6}$ $2x-6 \geq 0$, so $x \geq 3$.

c $h(x) = \frac{x+1}{x^2-9}$ All real x with $x \neq 3$ and $x \neq -3$.

d $k(x) = \sqrt[3]{x}$ All real numbers.

- 2 Find the domain of $f(x) = \frac{\sqrt{5-x}}{x-1}$. Write your answer in interval notation.

We need $5-x \geq 0$, so $x \leq 5$, and $x \neq 1$. Domain: $(-\infty, 1) \cup (1, 5]$.

Domain and range from graphs

- 3 The graph of $y = x^2 - 2$ is shown. State the domain and range of the function.

Domain: all real numbers. Range: $y \geq -2$.

- 4 The graph of $y = \sqrt{x+3}$ is shown. State the domain and range of the function.

Domain: $x \geq -3$. Range: $y \geq 0$.

Finding the range algebraically

- 5 By completing the square, find the range of $f(x) = x^2 + 4x + 1$.

$f(x) = (x+2)^2 - 3$. Since $(x+2)^2 \geq 0$, the range is $y \geq -3$.

- 6 Consider $f(x) = -2(x-1)^2 + 5$.

a State the coordinates of the vertex. Vertex: $(1, 5)$.

b Does the parabola open upward or downward? Downward, since the leading coefficient $-2 < 0$.

c State the range of f . $y \leq 5$.

- 7 Find the range of $g(x) = \sqrt{9-x^2}$ (the upper half of a circle).

The domain is $[-3, 3]$; g increases from $g(\pm 3) = 0$ to a maximum $g(0) = 3$. Range: $0 \leq y \leq 3$.

Domain and range in applied contexts

- 8 A rectangular garden has perimeter 40 m. If x is the width, the length is $20 - x$, and the area is $A(x) = x(20 - x)$.
- a State the domain of A in this context (both dimensions must be positive). $0 < x < 20$.
- b Find $A(6)$ and $A(15)$. $A(6) = 6(14) = 84 \text{ m}^2$; $A(15) = 15(5) = 75 \text{ m}^2$.
- c State the range of A over this domain, to the nearest whole number. $A(x) = -(x - 10)^2 + 100$ has maximum 100 at $x = 10$ and approaches 0 at the endpoints: range $0 < y \leq 100$.
- 9 An open-top box is formed from a 12 cm square by cutting squares of side x from each corner, giving volume $V(x) = x(12 - 2x)^2$.
- a State the domain of V in this physical context. $0 < x < 6$.
- b Find $V(1)$ and $V(4)$. $V(1) = 1(10)^2 = 100 \text{ cm}^3$; $V(4) = 4(4)^2 = 64 \text{ cm}^3$.

Domain of composite and combined functions

- 10 Let $f(x) = \sqrt{x - 2}$ and $g(x) = \frac{1}{x}$. Find the domain of:
- a $(f + g)(x)$ Need $x \geq 2$ and $x \neq 0$: domain $x \geq 2$. b $(f \circ g)(x)$ $f(g(x)) = \sqrt{\frac{1}{x} - 2}$, needs $\frac{1}{x} \geq 2$ and $x \neq 0$: domain $0 < x \leq \frac{1}{2}$.
- 11 Find the domain of $f(x) = \frac{1}{\sqrt{x^2 - 4}}$, writing your answer in interval notation.
Need $x^2 - 4 > 0$ (strictly, since it's in a denominator under a root): $x < -2$ or $x > 2$. Domain: $(-\infty, -2) \cup (2, \infty)$.

Domain from rational functions with factoring

- 12 Find the domain of $f(x) = \frac{x^2 - 4}{x^2 - x - 6}$, factoring the denominator first.
 $x^2 - x - 6 = (x - 3)(x + 2)$. Domain: all real x with $x \neq 3$ and $x \neq -2$.
- 13 Find the domain of $g(x) = \frac{\sqrt{x + 4}}{x - 1}$, combining a root restriction with a denominator restriction.
Need $x + 4 \geq 0 \Rightarrow x \geq -4$, and $x \neq 1$. Domain: $[-4, 1) \cup (1, \infty)$.

Range using transformations

- 14 The range of $f(x) = x^2$ is $y \geq 0$. State the range of $g(x) = f(x) - 5 = x^2 - 5$ and of $h(x) = -f(x) = -x^2$, using transformation reasoning.
 $g(x) = x^2 - 5$ shifts the range down 5: range $y \geq -5$. $h(x) = -x^2$ reflects the range over $y = 0$: range $y \leq 0$.

- 15** The range of $f(x) = \sqrt{x}$ is $y \geq 0$. State the range of $g(x) = 3\sqrt{x} + 2$.
Multiplying by 3 scales the range but preserves $y \geq 0$; adding 2 shifts up: range $y \geq 2$.
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Domain and range of piecewise functions

- 16** A function is defined by $f(x) = x^2$ for $x < 0$ and $f(x) = 2x + 1$ for $x \geq 0$. State the domain and range.
Domain: all real numbers (every x has a defined piece). For $x < 0$: $f(x) > 0$ (excluding 0 itself, approaching it). For $x \geq 0$: $f(x) \geq 1$. Combined range: $y > 0$.